

TOOL INSTALLATION COMMANDS

```
sudo apt update
sudo apt install build-essential git python3 python3-pip
sudo apt install iverilog gtkwave
sudo apt install yosys
sudo apt install magic
sudo apt install netgen
sudo apt install klayout
git clone https://github.com/The-OpenROAD-Project/OpenLane.git
cd OpenLane
make setup
sudo apt install docker.io
sudo usermod -aG docker $USER
```

RTL TO GDS FLOW

- mkdir new // **create folder named new**
- nano code.v
- nano code_tb.v

SIMULATION

- iverilog -o sim.out code.v code_tb.v // **Compiles the design and testbench**
- vvp sim.out // **Generates waveform file(.vcd)**
- gtkwave apb.vcd // **Opens waveform viewer**

Close the waveform viewer

SYNTHESIS

- yosys // **Open Yosys tool**
 - read_verilog code.v // **Read the design code**
 - synth -top apb // **Converts RTL into logic gate**
 - dfflibmap -liberty /home/reni/OpenROAD-flow-scripts/flow/platforms/sky130hd/lib/sky130_fd_sc_hd_tt_025C_1v80.lib
// **Maps flip-flops to standard cell library**
 - abc -liberty /home/reni/OpenROAD-flow-scripts/flow/platforms/sky130hd/lib/sky130_fd_sc_hd_tt_025C_1v80.lib
// **Optimizes logic using standard cells**
 - write_verilog mapped.v // **Generates gate-level netlist**
 - show // **View Netlist**
 - exit

PHYSICAL DESIGN

- cd ~/OpenLane
- mkdir -p designs/apb/src
- cp ~/new/mapped.v designs/apb/src/ // **Moves synthesized netlist into OpenLane design folder**
- nano ~/OpenLane/designs/apb/config.tcl
- make mount
- ./flow.tcl -interactive // **Starts OpenLane environment**
 - prep -design apb -overwrite // **Prepares design**
 - run_floorplan

- openroad -gui // **Open the design**

tcl window will open ,in that window type the following commands

- ❖ read_lef designs/apb/runs/RUN_2026.04.05_10.35.41/tmp/merged.nom.lef
- ❖ read_def designs/apb/runs/RUN_2026.04.05_10.35.41/results/floorplan/apb.def

close the tcl

- run_placement
 - openroad -gui
 - ❖ read_lef designs/apb/runs/RUN_2026.04.05_10.45.43/tmp/merged.nom.lef
 - ❖ read_def designs/apb/runs/RUN_2026.04.05_10.45.43/results/placement/apb.def
 - run_cts
 - openroad -gui
 - ❖ read_lef designs/apb/runs/RUN_2026.04.05_10.45.43/tmp/merged.nom.lef
 - ❖ read_def designs/apb/runs/RUN_2026.04.05_10.45.43/results/cts/apb.def
 - run_routing
 - openroad -gui
 - ❖ read_lef designs/apb/runs/RUN_2026.04.05_10.45.43/tmp/merged.nom.lef
 - ❖ read_def designs/apb/runs/RUN_2026.04.05_10.45.43/results/routing/apb.def
 - run_magic // **Generates gds**
 - run_magic_drc // **Check for DRC**
 - run_magic_spice_export // **Extracts netlist from layout**
 - run_lvs // **Check LVS**
 - exit
 - exit
- openroad
- read_liberty <lib_path>
 - read_db designs/apb/runs/<RUN>/results/routing/apb.odb
 - report_power // **Calculate Power**
 - report_design_area // **Report Area**
- klayout designs/apb/runs/<RUN>/results/signoff/apb.gds // **Open Layout**